

Wenbo Lyu

B.Sc. Graduate

Curriculum Vitae

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Education and qualifications

B.Sc. In Geographic Information Science Shaanxi Normal University 2025

Research Experience

2026.6- Research Assistant, The Hong Kong Polytechnic University
2025.9-2026.6 Research Assistant, Eastern Institute of Technology, Ningbo
2024.8-2025.8 Research Intern, The Hong Kong University of Science and Technology (Guangzhou)

Research interests

- My research interests lie in **advancing methodologies in spatial causal inference** and **developing high-performance computational tools**, with a primary focus on *R packages*.
- Currently, my work centers on **Empirical Dynamic Modeling (EDM)** framework for modeling *dynamic system*, **information theory** for quantifying *information flow* and *variable interactions*, **ordinary differential equations (ODEs)** for characterizing *spatiotemporal processes* and *system evolution*, as well as **counterfactual** and **potential outcomes** framework for estimating *causal effects*. I am particularly interested in leveraging these approaches to address critical challenges in *urban sustainability*, *climate change mitigation*, and broader global issues.
- I specialize in *data analysis*, *statistical modeling*, and developing open-source analytical tools, including *R packages*, using **R**, **C++**, and **Python**, with a strong focus on *spatial analysis*. I actively contribute to the R geospatial community and am dedicated to advancing open-source geospatial software.

Selected research papers

1. Lyu, W, Dai, S, Song, Y, Zhao, W, Yi, W, Xiao, Y, and Jia, N (2026). Measuring causal strengths from spatial cross-sectional data with geographical cross mapping cardinality. *International Journal of Geographical Information Science*, 1–23. DOI: 10.1080/13658816.2026.2687121.
2. Lyu, W, Lei, Y, Yi, W, Song, Y, Li, X, Dai, S, Qin, Y, and Zhao, W (2026). Causal discovery in urban data with temporal empirical dynamic modeling: The R package tEDM. *Computers, Environment and Urban Systems* 127, 102435. DOI: 10.1016/j.compenvurbsys.2026.102435.
3. Lv, W, Lei, Y, Liu, F, Yan, J, Song, Y, and Zhao, W (2025). gdverse: An R Package for Spatial Stratified Heterogeneity Family. *Transactions in GIS* 29(2), e70032. DOI: 10.1111/tgis.70032.
4. Lv, W, Liu, F, Cai, K, Cao, Y, Deng, M, Liang, W, Yan, J, and Wang, G (2024). Distinguishing the Impacts and Gradient Effects of Climate Change and Human Activities on Vegetation Cover in the Weihe River Basin, China. *Journal of Geophysical Research: Biogeosciences* 129(10), e2024JG008297. DOI: 10.1029/2024jg008297.
5. Li, B, Lv, W, Ding, H, Song, Y, and Zhao, W (2026). New-type urbanization policy, urban expansion, and employment quality in chinese cities: evidence from a quasi-natural experiment in China. *GIScience & Remote Sensing* 63(1), 2716413. DOI: 10.1080/15481603.2026.2716413.
6. Xu, L, Shi, H, Lyu, W, Zhang, Y, and Zhao, H (2026). Beyond correlation: Unveiling the causal link between street-scapes and recreational walking/cycling. *Journal of Transport Geography* 136, 104789. DOI: 10.1016/j.jtrangeo.2026.104789.
7. Chen, C, Song, Y, Lv, W, Shemery, A, Hampson, K, Yi, W, Zhong, Y, and Wu, P (2025). Predicting pavement cracking performance using laser scanning and geocomplexity-enhanced machine learning. *Computer-Aided Civil and Infrastructure Engineering*. DOI: 10.1111/mice.13489.

Note: Publications under “Lyu, W.” and “Lv, W.” both refer to Wenbo Lyu.

Selected awards and honours

2024 Longi Non-Education Major Scholarship
2024 First Prize in the 13th National University Student GIS Application Skills Competition
2023 Grand Prize in the 12th National University Student GIS Application Skills Competition
2023 First Prize in the Second National University Student Ecological Environment Management Research Innovation Competition
2023 Second Prize of the 5th 'Guodi Cup' National College Student Natural Resource Science and Technology Competition
2021 Outstanding Individual in Military Training Publicity for College Students, Shaanxi Normal University